Unit 2.1 Printables

DNA to Proteins (HS-LS1-1) · 8-day version: Siamese Cats

Everything students write goes in the notebook (Unit_2.1_Student_Notebook.pdf). These pages are only the extras.

Day	Print	Copies
1, 3, 4, 7	Glue-in strips (page 2): Day 1 key ideas, Day 3 base key and code table, Day 4 code table, Day 7 Write It data	1 per student (cut apart)
3	Gummy DNA Lab directions (page 3) and mats (page 4)	1 directions page per student, 1 mat set per pair
4	Card sort, cards A to F (page 5)	1 set per group

Day 3 lab materials (per pair)

- 2 licorice strands (6 to 8 inches)
- 24 candy bases: 6 red (A), 6 yellow (T), 6 green (C), 6 orange (G), plus a few spare
- Toothpicks, a paper plate
- No food? Use colored beads and pipe cleaners, or colored paper circles

Cut on the dashed lines. Strips are sized to fit half of a notebook page.

Glue-in strips

Cut apart on the dashed lines and glue into the notebook.

Day 1 · Part 2: Key ideas

- **DNA** in every cell stores instructions.
- A **gene** is a section of DNA with the instructions for one **protein**.
- **Proteins** do jobs for the cell.
- A kitten gets one set of chromosomes (long DNA molecules with many genes) from each parent.

Day 3 · glue in above Part 1: Lab record

Red = A Yellow = T Green = C Orange = G

DNA triplet	Amino acid	DNA triplet	Amino acid
ATG	Met	AGA	Arg
GGA	Gly	CTG	Leu
TTT	Phe	GAC	Asp

Day 4 · glue in at Part 2: Card sort

DNA triplet	Amino acid	DNA triplet	Amino acid
GGA	Gly	AGA	Arg
GAG	Glu	GTG	Val
GAA	Glu	CTG	Leu
CCG	Pro		

Day 7 · Part 3: Write It data, two Siamese kittens

	Kitten raised warm	Kitten raised cool
Ear skin temperature (°C)	34	26
Ear fur color	Light tan	Dark brown
Back skin temperature (°C)	36	35
Back fur color	Cream	Cream

Day 3 · Gummy DNA Lab: Directions

Question: How does a cell's copy of a gene become a chain of amino acids, and what happens if one letter of DNA changes?

You will build a short model of the start of a cat's gene for a pigment-making protein. (The letters are an illustration, not the real gene.) Record everything in your notebook, Part 1 and Part 2 on the Day 3 pages.

Materials (per pair)

- 2 licorice strands (the backbone: sugars and phosphates)
- Candy bases: 6 red (A), 6 yellow (T), 6 green (C), 6 orange (G), plus 2 extra of each
- Toothpicks, a paper plate, the mats, the base key and code table

Safety: Keep materials clean. Do not eat the lab materials until your teacher says you may. Toothpicks are sharp. Tell your teacher about any food allergy.

Part A: Build the template strand (6 min)

1. Attach the bases to one licorice strand, left to right: **T A C C C T G A C A A A**
2. Write the colors you built in your notebook.

Part B: Build the partner strand and make the ladder (8 min)

1. Build the partner strand on the second licorice strand. Pair each base: A with T, C with G.
2. Join the strands with toothpicks. Gently twist the ladder. Have your teacher check it.
3. Which strand carries the coding letters? How do you know? Answer in your notebook.

Part C: Copy the message, then read it (10 min)

The cell does not use the DNA directly to make a protein. It **copies the gene's message** from the template strand, and then the **copy is read** to link amino acids. The DNA stays safe.

1. Pull the two strands apart. Keep the template strand.
2. Build a new strand by pairing with the template. This is the **copy**. Write its sequence.
3. Read the copy three letters at a time with the code table. Write the amino acid chain.
4. Line up one bead or paper tag per amino acid on your plate. This chain would fold to make a protein.
5. Explain the path in your own words, starting with the DNA.

Part D: Change one letter (7 min)

1. On your template strand, replace the 4th base (the first C of the second triplet) with a T. Swap a green gummy for a yellow one. The template now reads **T A C T C T G A C A A A**.
2. Build the new copy by pairing with the changed template. Write its sequence.
3. Read it with the code table. Write the amino acid chain.
4. Compare with Part C. How many amino acids changed? Which one?
5. Predict: could a protein built from this changed chain work differently? Explain.

Connect to the Siamese cat

1. In a Siamese cat, a one-letter change in the gene for a pigment-making protein swaps one amino acid. What did your one-letter change do to the chain in the model?

2. Name two ways your gummy model is different from real DNA.

Clean up: Put toothpicks and candy in the trash or the labeled container. Wipe your table. Wash your hands.

Day 3 · Gummy DNA mats

Slide the licorice strand along the mat. One base per box, three boxes per amino acid.

Parts A and B: template strand (T A C C C T G A C A A A) and partner strand

Template

Partner

Part C: the copy

Amino acids: _____ – _____ – _____ – _____

Part D: changed template (T A C T C T G A C A A A) and its copy

Amino acids: _____ – _____ – _____ – _____

Day 4 · Card sort

Cut on the dashed lines. Sort into three categories, glue into the notebook, and write your reasons.

A

In a Siamese cat, one letter changes in the gene for a pigment-making protein: GGA becomes AGA.

B

A gene's triplet changes from GAG to GAA.

C

A gene's triplet changes from CTG to CCG.

D

One letter changes in a stretch of DNA that has no gene and no switch.

E

One letter changes in a switch next to a gene so the gene can never be turned on, and its protein is not made.

F

In a human gene, GAG becomes GTG.